

Experimental Protocol

Water Resistance Test

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Objective of the Experiment

Since the developed enclosure is intended to be buried approximately 5 cm below the surface of the ground in its application environment—primarily to protect it from vandalism and damage caused by mowing or maintenance activities around the trees—a verification of its water resistance is necessary. The objective of this experiment is therefore to evaluate the effectiveness of the sealing measures and to assess the suitability of the 3D printing process with respect to water resistance.

Experimental Setup

To create a controlled and reproducible environment, the experimental setup is designed in such a way that all investigations can be conducted under standardized laboratory conditions. The aim is to simulate the underground installation scenario as realistically as possible, as expected during practical deployment.

The experiments are conducted using plastic containers with a volume of 20 liters each, which are filled with soil. In order to accurately reflect the typical regional soil conditions, the soil used was collected directly from the campus grounds of Flensburg University of Applied Sciences. It consists of a natural mixture of sand, stones, and humus and is therefore considered representative of the soil types commonly found in Northern Germany.

After filling the containers, the soil is compacted uniformly using a body weight of approximately 85 kg, in order to realistically simulate the soil compaction that typically occurs during burial. For this purpose, the body weight of a team member was practically employed, who compacted the soil step by step by firmly stepping on it.

Before the actual test series begins, the natural moisture content of the soil is determined. For this purpose, an SMT100 sensor is used, which accurately measures the volumetric water content in the soil. The measured values serve as a reference for the subsequent controlled wetting.

In the first test series, the impact of a heavy rainfall event on the buried enclosure is to be simulated. For this, the maximum documented rainfall in the Flensburg region was analyzed, which is approximately 40 liters per square meter [1] [2]. This value was proportionally scaled to the base area of the 20-liter containers used, resulting in a

target amount of 4.4 liters of water per container. This corresponds to approximately 22% of the container's volume.

This quantity of water was applied in a controlled and uniform manner to the surface of the compacted soil to simulate a realistic scenario of intense rainfall. Subsequently, the containers were placed in a calm, vibration-free location in order to minimize external influences on the experiment and to ensure uniform conditions for water infiltration.

Execution of the Experiment

At the beginning of the experiment, the two enclosures to be tested were each placed centrally into standard commercial plastic containers with a capacity of 20 liters. Prior to filling the containers with soil, the cable inlets on the enclosures were carefully sealed to ensure that any potential water ingress could only occur via the enclosure's lid seal or through the material of the enclosure itself. This measure ensured that any observed leakage could be clearly attributed to the structural design rather than to improper sealing of technical components.

Subsequently, the containers were completely filled with soil, such that the enclosures were fully covered and positioned at an installation depth of 120 mm below the soil surface. This depth corresponds to the intended installation depth in the actual field application and ensures a realistic assessment of sealing performance under typical environmental conditions.

Following the filling process, the natural soil moisture content was determined using an SMT100 sensor. The initial measurement yielded a baseline value of approximately 2 vol%, classifying the soil as nearly dry.



Abbildung 1: Enclosure with inserted seal

Thereafter, the previously calculated amount of 4.4 liters of water was evenly distributed over the surface of the soil in order to realistically replicate the moisture conditions following a heavy rainfall event. After the water had fully infiltrated the soil, a second moisture measurement was conducted using the SMT100 sensor. This measurement revealed a volumetric soil moisture of 20.4%, indicating a significant increase in water content compared to the initial state.



Abbildung 2: SMT100 with microcontroller

To simulate the soil compaction typically occurring after the burial of technical components, the soil was compacted uniformly with a body weight of approximately 85 kg following the water application. This compaction was manually performed by a team member stepping firmly on the soil surface.

Finally, the containers were placed in a quiet, vibration-free location to minimize external influences during the saturation and observation phases. This ensured stable environmental conditions for the evaluation of the water resistance of the enclosures.



Abbildung 3: Buried enclosure with depth measurement



Abbildung 4: Soil-filled test containers under laboratory conditions

Results and Observations

Upon completion of the test series, the two enclosures were carefully removed from the test containers in order to examine their condition following defined exposure to moisture and soil pressure. The enclosures were opened immediately after removal to reliably detect any potential signs of water ingress.

Visual and tactile inspection of the interior spaces revealed that no water had penetrated the enclosures. The sealing measures—particularly the circular gasket attached to the lid—proved effective under the selected test conditions. Furthermore, no signs of moisture ingress were observed at the previously sealed cable inlets. Consequently, the simulated irrigation, which modeled a heavy rainfall event, did not result in any structural compromise of the enclosure's functionality.

In addition to the positive evaluation of water resistance, certain structural weaknesses were identified that require consideration with regard to practical deployment in soil. Specifically, it was observed that opening the lid of the enclosure while it is embedded in the ground is significantly hindered, as the enclosure cannot be adequately stabilized within the soil. As a result, it is difficult to generate the necessary counterforce when unscrewing the lid without unintentionally applying mechanical stress to the connected cables or plugs. Without additional fastening mechanisms for securing the housing, this process presents an increased risk to the integrity of the electrical components.



Abbildung 5: Excavated enclosure during visual inspection

Another critical point concerns the ingress of soil particles when opening the enclosure. It was observed that even minor movements of the lid can cause loose soil particles from the edge of the housing to fall into the interior, particularly in the absence of an additional cover or protective sleeve. This behavior poses a potential threat to the operational safety of the internal electronics, especially during prolonged outdoor use or under varying environmental conditions.

In summary, it can be concluded that the primary requirement of water resistance was successfully met under realistic conditions. At the same time, the conducted observations indicate a need for optimization regarding mechanical handling and protection against contamination during the opening process, in order to ensure reliable long-term use in outdoor environments.



Abbildung 6: Opened enclosure after moisture exposure test

Future Work

The conducted investigations have demonstrated that the developed enclosure provides sufficient water resistance under controlled laboratory conditions and is, in principle, suitable for the intended underground deployment. Nevertheless, the practical observations made during the experiments also revealed certain structural weaknesses that should be specifically addressed in future development cycles.

A key area for optimization lies in the mechanical handling of the enclosure when installed in the ground. As it is not possible to apply adequate counterforce when opening the lid without inadvertently exerting stress on the cable connections, the development of a suitable fixation mechanism is recommended. One possible solution would be the integration of an external support frame or an anchoring system in the soil, which stabilizes the enclosure during maintenance operations without jeopardizing the electrical connections. Alternatively, the implementation of low-torque locking mechanisms—such as a bayonet lock—could significantly facilitate the disassembly process.

Furthermore, the introduction of additional structural protective measures to prevent the ingress of soil particles during the opening process appears advisable. A surrounding dirt lip or an external sealing sleeve could help prevent loose soil from entering the interior of the enclosure when the lid is loosened. This would enhance the operational reliability of the internal electronics in the long term while also reducing the maintenance effort.

For future investigations, it is recommended that the test conditions be expanded to include more realistic environmental influences. These include temperature cycles, freeze-thaw transitions, mechanical stress due to fluctuating soil moisture, and microbiological effects. Long-term field tests under natural weather conditions could

also provide valuable insights into the durability and aging resistance of the materials and sealing systems used.

It should be noted, however, that the installation depth of 120 mm selected in this experiment was intentionally greater than the depth planned for the actual application. In the intended deployment scenario, the enclosure is to be installed only 50 mm below the soil surface. Under this condition, it can be assumed that both access to the enclosure and handling of the lid will be significantly easier, due to the substantially lower mechanical stress and reduced soil coverage. Thus, enclosure handling at the planned installation depth does not constitute a fundamental problem.

References

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